

Solution

HUMAN EYE -01

Class 10 - Science

Section A

1. Due to atmospheric refraction, apparent length of the day increases by 4 minutes.
2. The rays of light from upper and lower edges of Sun bend unequally. Due to unequal bending of light, the image of Sun appears oval and larger in size.
3. The twinkling of a the star is due to atmospheric refraction. Distant stars act like a point source of light. As the beam of starlight keeps deviating from its path, the apparent position of star keeps on changing because physical conditions of earth's atmosphere are not constant. Hence, the amount of light which enters our eyes fluctuates from bright to faint. This is the "Twinkling effect of star". But planets are much closer to us than stars, so they are not point-sized objects to our eye. Hence, the fluctuations have a negligible effect and they don't seem to twinkle.
4. In the absence of an atmosphere the sky appears black due to no scattering of light.
5. Yes ,the earth appears blue to an astronaut in a spaceship due to the large scattering of blue light by the atmosphere.
6. When a white light (from sun) enters the earth's atmosphere, it gets scattered away due to the atmospheric particles. Since, blue colour has the minimum wavelength, so blue colour scatters the most and thus the sky appears blue.
7. During sunset and sunrise, the rays have to travel a larger part of the atmosphere because they are very close to the horizon. Therefore, light other than red is mostly scattered away. Most of the red light, which is the least scattered, enters our eyes. Hence, the sun appears red.
8. The wavelength of red light is the longest in visible light, so the red color is scattered the least by air molecules of the atmosphere and therefore it can reach to a longer distance. That's why the red color is used for danger signal lights.

Section B

9.

(c) A, C and D

Explanation: The rainbow is a natural spectrum of sunlight appearing in the sky after a rain shower. It is formed due to the dispersion of sunlight by the tiny water droplet, present in atmosphere. Water droplets act like prism.

It refracts and disperse the incident sunlight, then reflect it internally (total internal reflection) and finally refract it again, when it emerges out of the water droplet. Red colour appear on top and violet at the bottom of the rainbow.

10.

(d) Atmospheric refraction of starlight

Explanation: The atmospheric refraction is responsible for twinkling of stars. As the light from the stars enters the earth's atmosphere it suffers refraction and cause the twinkling.

11.

(c) Refraction, dispersion and internal reflection.

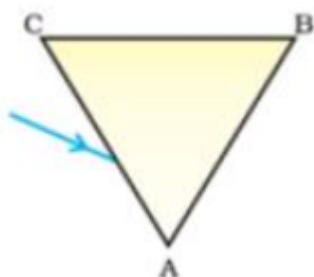
Explanation: A rainbow is a natural spectrum of seven colours appearing in the sky after a rain shower. It is caused by dispersion of sunlight by tiny water droplets, present in the atmosphere. The water droplets act like small prisms. They refract and disperse the incident sunlight, then reflect it internally, and finally refract it again when it comes out of the raindrop.

12. (a) Bifocal lens whose upper portion is concave lens and lower portion is convex lens

Explanation: The upper portion (concave lens) facilitates distant vision and the lower portion (convex lens) facilitates near vision.

13.

(c)



Explanation: When the prism is inverted, then after dispersion the colors which will be observed are VIBGYOR. So, the figure is the correct image to get the third color as the color of the sky i.e. Blue.

14.

(c) A concave lens of focal length -1.4 m and power -0.71 D

Explanation: In normal eye, the far point is infinity. The lens used should be such that an object at infinity forms virtual image at 1.4 m.

$$v = -1.4 \text{ m}, u = -\infty, f = ?$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \frac{1}{(-1.4)} - \frac{1}{(-\infty)} = -\frac{1}{1.4}$$

$$\therefore f = -1.4 \text{ m}$$

$$\text{Power, } P = \frac{1}{f} = \frac{1}{-1.4} = -0.71 \text{ D}$$

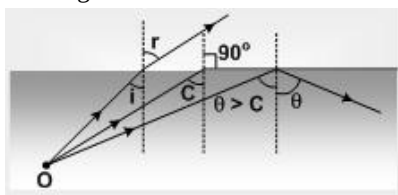
A concave or divergent lens of focal length 1.4 m and power -0.71 D should be used to restore proper vision.

15.

(c) light should pass from denser to rarer medium

Explanation: The two conditions for total internal reflection are:

- The ray of light must travel from a denser medium into a rarer medium.
- The angle of incidence in the denser medium must be greater than the critical angle for that pair of media.



16. (a) Concave lens

Explanation: Concave lens

17. (a) Pupil

Explanation: The pupil is the opening in the centre of the iris that allows light to flow through. The iris controls the quantity of light that enters the eye by adjusting the size of the pupil.

18. (a) 25 cm

Explanation: 25 cm

19.

(d) Refraction, dispersion and internal reflection

Explanation: Rainbow is formed due to the phenomena: Refraction, dispersion, and internal reflection due to falling of sunlight on rain droplets.

20. (a) The short-sighted eye, the correction of short-sightedness, the long-sighted eye and the correction of long-sightedness

Explanation: In short-sighted eye, image is formed in front of retina. To correct it, a concave lens of suitable focal length is used. In long-sighted eye, image is formed behind retina. To correct it, a convex lens of suitable focal length is used.

21.

(d) The band of 7 colours.

Explanation: When a white light is passed through a prism it gets split into combination of seven colours which is known as the spectrum. The colors of the spectrum of white light are those seen in a rainbow. They are usually named in order as: red, orange, yellow, green, blue, indigo and violet.

22.

(b) The lens allows the eye to focus on far objects as well as near objects.

Explanation: Ability of the eye lens to focus both near and distant objects by adjusting its focal length is called accommodation.

23.

(c) 1.414

Explanation: The sine of the critical angle is equal to the reciprocal of the refractive index of that material i.e.

$$\sin c = \frac{1}{\mu}$$

$$\sin 45^\circ = \frac{1}{\mu}$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\mu}$$

$$\mu = \sqrt{2} = 1.414$$

24. (a) Pupil

Explanation: Pupil

25.

(b) Rods

Explanation: Rods

26.

(b) The person can read letters from 6 m which the normal eye can read from 9 m

Explanation: The person can read letters from 6 m which the normal eye can read from 9 m

27.

(d) May see a primary and a secondary rainbow as concentric circles

Explanation: May see a primary and a secondary rainbow as concentric circles

28.

(d) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.

Explanation: The prism Y recombines the seven colours to form white light.

29.

(c) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

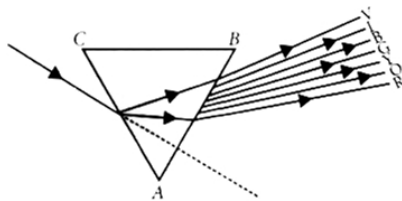
Explanation: (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

30. (a) Statement 1 is true but statement 2 is false.

Explanation: The range of vision for a normal eye is from 25 cm to infinity. The eye is able to see clearly all objects in front of it by adjusting its focal length suitably.

31. (a) Figure (ii)

Explanation: Out of the following orientations, figure (ii) represents the case in which, after dispersion, the third colour from the top is blue i.e. colour of the sky.



32.

(c) Dispersion

Explanation: Dispersion is the phenomena of splitting of white light into its constituent seven colours (VIBGYOR) on passing through a glass prism.

33.

(d) Presbyopia

Explanation: Presbyopia

34.

(b) A and B

Explanation: Both of these phenomena are due to atmospheric refraction. The temperature and density of different layers of atmosphere keep varying. When the light enters the earth's atmosphere it undergoes refraction continuously, due to changing refractive index i.e. light travels from rarer to denser medium and hence It bends towards the normal successively.

35.

(c) refraction of light by different layers of varying refractive indexes

Explanation: Stars twinkle due to atmospheric refraction of light by different layers of atmosphere which are having different refractive indexes.

36.

(b) opaque eye lens

Explanation: Explanation:

- i. A person suffering from cataract has an opaque eye lens.
- ii. In cataracts, the crystalline lens of the people becomes milky and cloudy at an old age.
- iii. In this condition, the person loses his/her eyesight partially or completely due to the cloudiness in front of the eyes.
- iv. The milky or cloudy condition makes him unable to see the thing clearly.
- v. Therefore the eye lens becomes opaque and a person can not see the objects clearly.

37. (a) light is least scattered

Explanation: light is least scattered

38. (a) red colour is least scattered.

Explanation: Red colour scattered the least when strikes by the small particle of fog and smoke because it has the maximum wavelength (visible spectrum). Hence at large distance also, we can see the red colour clearly.

39.

(c) Convex lens

Explanation: Convex lens

40.

(b) Only (i) and (ii)

Explanation: Diagram shows hypermetropia. This defect arises either because the focal length of the eye lens is too long or the eyeball has become too small.

41.

(c) ultraviolet light

Explanation: The retinal cones of bees are sensitive to ultraviolet light. Hence bees respond to ultraviolet light.

42.

(d) Violet

Explanation: Violet

43. (a) ciliary muscles

Explanation: The curvature of the eye lens changes as the ciliary muscles relax or contract. The curvature of the eye lens alters the focal length of the eyes. As a result, the activity of ciliary muscles causes a change in the focal length of an eye lens.

44.

(c) Red

Explanation: Red

45.

(c) Only (i) and (ii)

Explanation: Advance sunrise and delayed sunset are due to atmospheric refraction.

46. (a) Violet and blue lights get scattered more than lights of all other colours by the atmosphere.

Explanation: The clear sky is blue in color because blue light is scattered more than other colour of light by molecules of air.

47.

(b) scattering of light is not enough at such heights.

Explanation: Scattering of light takes place because of the particles present in the atmosphere. At high altitude due to the absence of atmospheric scattering of light do not take place and hence sky appears dark to passengers flying at high altitude.

48.

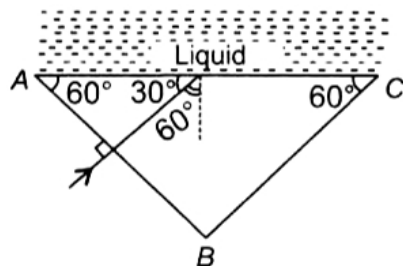
(c) reflection of sky in water

Explanation: The bluish color of water in the deep sea is due to the scattering of light because blue color has the smallest wavelength than other colors and therefore, due to the scattering of water particles by Sun's radiations, the color of the water is blue in the deep sea.

49. (a) Atmospheric refraction

Explanation: Twinkling of stars is due to atmospheric refraction. Distant star acts like a point source of light. When the starlight enters the earth's atmosphere it undergoes refraction continuously, due to changing refractive index i.e. from Rarer to denser medium. It bends towards the normal successively, hence the amount of light enters our eyes fluctuates sometimes bright and sometimes faint.

50. (c) Look towards the raindrops with the light source behind him
Explanation: Look towards the raindrops with the light source behind him
51. (c) remains unchanged
Explanation: There is no change to the image distance in the eye when we increase the distance of an object from the eye. To see closer or distant object clearly, the eyes, due to its ability of accommodation, can increase or decrease focal length of its lens, so that the image is always formed at retina.
52. (d) Is scattered the least by smoke or fog
Explanation: The red color of the danger signal installed at the top of a tall building can be easily seen from a distance than other colors because red color, having the longest wavelength, is scattered least by smog or fog than other colors and that's why it is visible from a distance.
53. (d) Refraction
Explanation: Refraction
54. (c) Cornea
Explanation: Cornea
55. (a) Combined effect of dispersion, refraction and reflection.
Explanation: Rainbow is an example of the dispersion of sunlight by the water drops in the atmosphere. This is a phenomenon due to combined effect of dispersion, refraction and reflection of sunlight by spherical water droplets of rain.
56. (a) Black
Explanation: The sky appears blue due to the scattering of the blue colour by the earth's atmosphere. In moon there is no atmosphere hence nothing to scatter light. Thus, the sky appears dark or black as seen from the moon surface.
57. (b) Short-sightedness or myopia
Explanation: Short-sightedness or myopia
58. (a) Scattering of light
Explanation: At Sunrise or Sunset, the reddish appearance of Sun is due to the scattering of light, and since Red color has the highest wavelength and is scattered least.
59. (d) $\mu < \frac{3\sqrt{3}}{4}$
Explanation: Total internal reflection takes place when angle of incidence is greater than critical angle.



Here, $\mu_l = \mu, \mu_g = \frac{3}{2}$

We know,

$$\sin C = \frac{\mu_l}{\mu_g} = \frac{\mu}{3/2} = \frac{2\mu}{3}$$

Light suffers total internal reflection, $i > C$ or $60^\circ > C$

$$\Rightarrow \sin 60^\circ > \sin C \Rightarrow \frac{\sqrt{3}}{2} > \frac{2\mu}{3}$$

$$\therefore \mu < \frac{3\sqrt{3}}{4}$$

60. (a) $\lambda_v < \lambda_y < \lambda_r$

Explanation: Colors and corresponding wavelengths of visible spectrum.

Color	Wavelength (nm)
Violet	380-450
Blue	450-475
Cyan	476-495
Green	495-570
Yellow	570-590
Orange	590-620
Red	620-750

61.

(d) Real and inverted

Explanation: Real and inverted

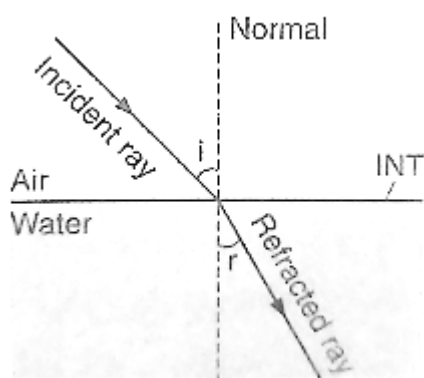
62.

(a) By a glass prism but not by rectangular glass slab.

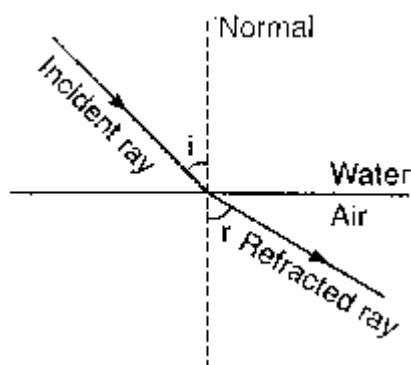
Explanation: The angle of deviation through a triangular prism is the angle between the incident ray and the emerging ray (angle δ). However, in glass slab, the incident ray and the emergent ray are parallel to each other thus angle of incidence is equal to the angle of emergence.

Section C

63. a. **From air to water :** Since the light is going from rarer (air) to denser (water) medium, therefore it will bend away from normal after refraction.



- b. **From water to air :** Since the light is going from denser (water) to rarer (air) medium, therefore it will bend towards normal after refraction.

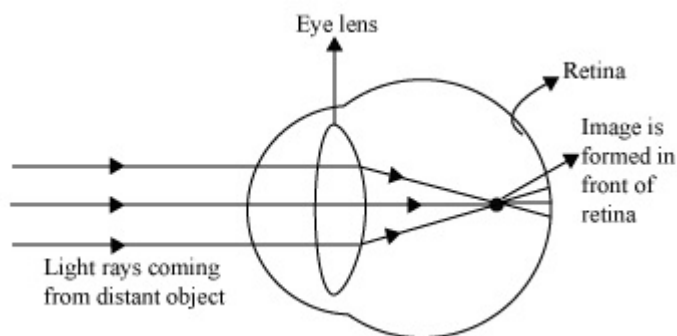


64. We can see distant and near objects clearly because of the power of accommodation of the eye. Accommodation is the name given to the ability of the lens of the eye to change its focal length and produce focused images of both distant and near objects on the retina.

Ciliary muscle helps in changing the curvature of the lens. They are attached to the lens and contract or release to change the lens shape and curvature.

Blind spot is an area where a person's view is obstructed. It is the small, circular, optically insensitive region in the retina where fibres of the optic nerve emerge from the eyeball. It has no rods or cones. No image is formed at the blind spot because there are no light sensitive cells called photoreceptors, i.e., cones and rods to detect the light and send it to the brain.

65. a. Myopia or Short-sightedness is the inability of an eye in viewing long distance objects. The image in this case falls before the retina.
 b. A myopic eye has its far point nearer than infinity. It forms the image of a distant object in front of its retina as shown in figure.



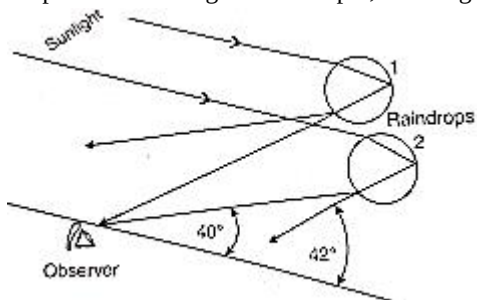
- c. For every myopic eye, there exists a far point beyond which clear image cannot be seen. Short-sightedness is caused due to
 i. excessive curvature in cornea.
 ii. elongation of eyeball.

66. The conditions for observing a rainbow are:

- The sun comes out after a rainfall.
- The observer stands with the sun towards His/her back.

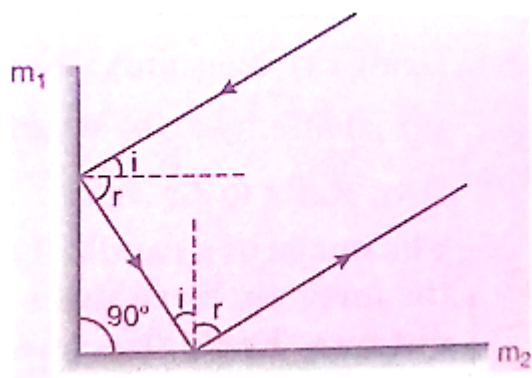
Formation of a rainbow :

The rays of light reach the observer through a refraction, followed by a reflection and a refraction. Figure shows red light, from drop 1 and violet light from drop 2, reaching the observer's eye.



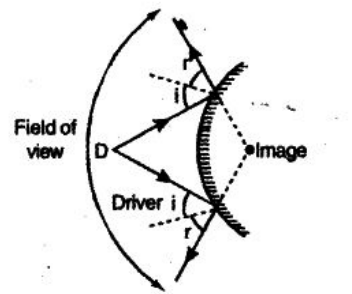
67. When two plane mirrors are placed at right angles to each other then, the incident and reflected rays will always be parallel to each other, whatever may be the angle of incidence.

Consider a plane mirror m_1 and m_2 placed at right angle to each other as shown in ray diagram. The incident and reflected rays are parallel to each other.



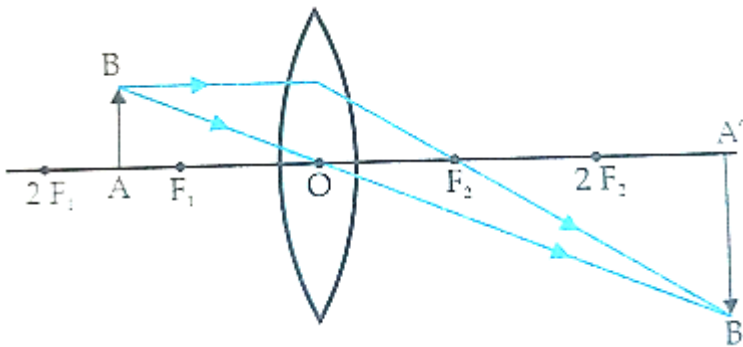
68. A convex mirror is used as a driver's mirror because the image produced by it is erect and diminished. As the size of the image is small so the field of view is wider. Moreover, the nature of image remains virtual, erect and diminished irrespective of the position

of the object. The view shown by a convex mirror is as shown in the figure below.



It is clear from the figure that the view produced by a convex mirror is wider

69. The position and nature of the image formed by a lens depends upon the position of object. Since, the object is placed between focus and centre of curvature of convex lens, Therefore image will be formed on other side of lens. A real, inverted and magnified image beyond $2F_2$ will be formed. The ray diagram is shown below.



70. The person is suffering from an eye defect called myopia. In this defect, the image is formed in front of the retina. Hence, a concave lens is used to correct this defect of vision.

Here Distance of object, $u = \text{infinity}$

Distance of image, $v = -80 \text{ cm}$

Let the focal length = f .

Therefore, by the lens formula we have,

$$-\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{-\infty} + \frac{1}{-80} = \frac{1}{f}$$

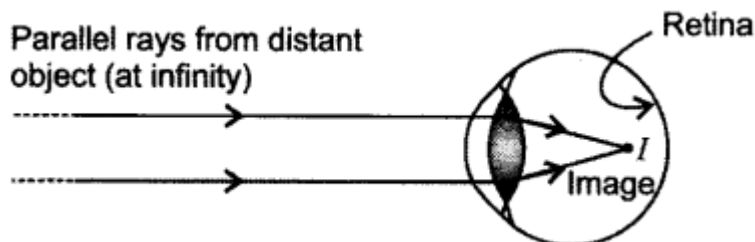
$$\Rightarrow f = -80 \text{ cm} = -0.8 \text{ m}$$

$$\text{Now, Power of lens, } P = \frac{1}{f} = \frac{1}{-0.8 \text{ in m}} = -1.25 \text{ D}$$

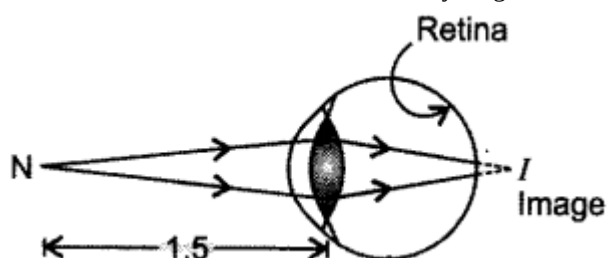
Since, The power of lens = -1.25 D . Therefore a concave lens of power -1.25 D is required by the person to correct this defect.

71. i. **When student is seated at the back seat:** Since a student at the back bench in a class is not able to see what is written on blackboard, Therefore, student is suffering from myopia eye defect. Myopia or Short-sightedness is caused due to

1. excessive curvature in cornea.
2. elongation of eyeball. Ray diagram is shown below.

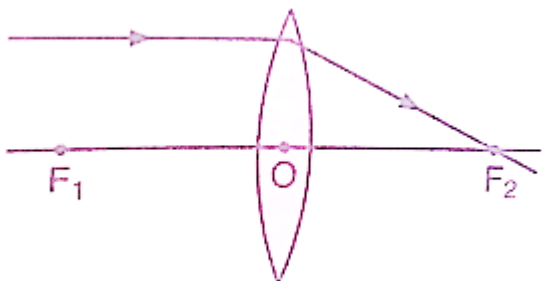


- ii. **When student is seated at front seat:** The ray diagram is shown below:



72. a. An object is seen only when the light illuminating it is reflected and reach our eye. According to Rayleigh's scattering Law, the intensity of scattered light $\propto \frac{1}{\lambda^4}$. Since yellow colour has small wavelength than red colour, so scattering of yellow colour is more than the red colour. It means signal is illuminated more brightly by the red colour than the yellow colour. Hence, the signals can be seen easily from a longer distance. Therefore, red colour is used for danger signals than yellow colour.
- b. In the visible spectrum, violet light have the shortest wavelength. Therefore, it has the highest refractive index. Therefore, when white light is passed through the prism, it split into its seven colours to give visible spectrum and violet colour having the shortest wavelength deviated the most.
73. Since, Ray of light from the object is parallel to the principal axis of convex lens. Therefore, the ray of light after refraction through the convex lens passes through the principal focus on the other side of the lens.

The Ray diagram is shown below:



74. The brightness of the image in the camera is:
- directly proportional to time of exposure(t).
 - directly proportional to the square of diameter of aperture of the lens system (i.e. light gathering power of the objective).
 - inversely proportional to the square of focal length of the lens system.
 - is inversely proportional to the square of the lateral magnification. i.e. Image Brightness $\propto (N_A/M)^2$; where N_A is the objective numerical aperture and M is the magnification.
75. Here, distance of far point, $x = 1.2\text{m}$
 For viewing distant objects, focal length of corrective lens, $F = -x = -1.2\text{ m}$
 We know that, Power of lens (P) is inversely proportional to the focal length (f).
 Therefore, $P = \frac{1}{f} = \frac{1}{-1.2} = -0.83\text{D}$.
 Therefore, the person should use the diverging concave lens of power -0.83D.